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MILITARY SPECIFICATION

COUPLER, ANTENNA, AN/SRA-33, FOUR-CHANNEL, AUTOMATICALLY TUNED

1. SCOPE

1.1 This specification covers an automatically tuned four-channel antenna coupler used to couple a group of transmitters, receivers, receiver-transmitters, or combinations thereof to a common antenna for operation in the frequency range 225.00 to 399.95 megahertz (MHz).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (HIGH-IMPACT), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-Q-9858	- Quality Program Requirements
MIL-M-10304	- Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore: General Specification
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-167	- Mechanical Vibration of Shipboard Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-785	- Reliability Program for Systems and Equipment Development and Production

HANDBOOKS

MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NAVSHIPS 94501 - Reliability Design Handbook

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE
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3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Material. Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

3.3 General requirements. The equipment covered in this specification shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.3.1 Environmental requirements.

3.3.1.1 Temperature. The equipment shall conform to the temperature requirements of class 1 of MIL-E-16400.

3.3.1.2 Enclosures. The antenna couplers and control-power supply shall be watertight. The Antenna Coupler Group, that is, AN/SRA-33, shall be splashproof.

3.3.1.3 Shock and vibration.

3.3.1.3.1 Shock. The equipment shall be tested on the medium weight shock test machine as class II equipment in accordance with MIL-S-901.

3.3.1.3.2 Vibration. The antenna coupler shall be tested in accordance with type I of MIL-STD-167.

3.3.1.4 Humidity. The humidity test as specified in MIL-E-16400 shall be required.

3.3.2 Cotton canvas and webbing. Cotton canvas and webbing shall not be used in this equipment.

3.3.3 Performance. Performance shall be reliable under conditions of intermittent or continuous operation upon exposure to the adverse conditions normally encountered in the Naval communication services, particularly in those applications for which the equipment is specifically intended. The adverse conditions to be considered shall include the temperature requirements of 3.3.1.1.

3.3.4 Time meters. Elapsed time meters are not required.

3.3.5 Synchros. Synchros are not required.

3.4 Detail requirements.

3.4.1 AN/SRA-33. The AN/SRA-33 shall consist of the following:

Quantity	Unit
1	Coupler, Antenna CU-1131/SRA-33
1	Coupler, Antenna CU-1132/SRA-33

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Quantity	Unit
1	Coupler, Antenna CU-1133/SRA-33
1	Coupler, Antenna CU-1134/SRA-33
1	Control-Power Supply C-4586/SRA-33
1	Cabinet, Electronic Equipment CY-3852/SRA-33
2	Cable Assembly, Special Purpose Electrical (Short) CX-8661/U
2	Cable Assembly, Special Purpose Electrical (Long) CX-8661/U

3.4.2 Equipment. The equipment shall be a four-channel, UHF, passive, antenna-coupler group, consisting of four antenna couplers, a control-power supply, an electronic equipment cabinet plus interconnecting cables. Each antenna coupler shall consist of a pair of tunable coaxial cavities, connected in tandem. Each antenna coupler shall be capable of being pretuned to any one of nineteen different preset frequencies, giving a total of 76 available preset frequencies for the four channels which may be selected remotely from four different control stations. The radio frequency (rf) input of each antenna coupler shall be brought out to a coaxial connector at the front panel of the equipment. The rf output connection shall lead to a common junction, which in turn shall be brought out to a coaxial connector on the rear of the equipment.

3.4.3 Radio frequency power. The equipment shall be capable of accepting rf power from all four input channels simultaneously, each channel having a carrier power level of not less than 100 watts and not to exceed 140 watts, modulated 80 percent with a 1-kilohertz (kHz) tone (see 4.4.2.2).

3.4.4 Electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility. The antenna shall operate within the electromagnetic interference emission, susceptibility, and equipment and systems self-compatibility limits in accordance with class 1A of MIL-STD-461, except that tests 01(T), CS 07, and CS 08 shall not be required.

3.4.5 Impedance. Nominal impedance exhibited to the antenna terminal and to each equipment terminal shall be 50 ohms.

3.4.6 Voltage standing wave ratio (VSWR). Maximum VSWR exhibited to a 50-ohm transmission line at the antenna terminal and at each equipment terminal shall be no greater than 2.0 to 1 (see 4.4.2.3).

3.4.7 Frequency range. Each channel shall be capable of being tuned to any frequency in the 225.00 to 399.95 MHz frequency band. Tuning over the band shall be in 0.05 MHz increments (see 4.4.2.4).

3.4.8 Tuning. The antenna coupler group shall also provide 19 pretuned frequencies per channel and a means of selecting any of the 76 pretuned frequencies of the antenna coupler group from four different control stations. The automatic tuning mechanism shall consist of a position type servo mechanism in conjunction with an autopositioner driven memory drum.

3.4.9 Manual tuning. Selection of any frequency in 225.00 to 399.95 MHz frequency band (0.05 MHz increments) shall be provided for by three manual selector switches located on the front panel, that is, a 10.0 MHz selector, a 1.0 MHz selector, and a 0.05 MHz selector.

3.4.10 Local preset tuning. A selector switch shall be provided which will select any of the 19 preset frequencies from the front panel.

3.4.11 Remote preset tuning. Provisions shall be made to provide for selection of any of the 19 preset frequencies from a remote operating position.

3.4.12 Manual-local-remote switch. A switch shall be provided on the front panel which will allow the desired mode of tuning to be selected. That is, manual tuning, local preset tuning, or remote tuning.

3.4.13 Tuning time. Maximum tuning time shall not exceed 15 seconds.

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3.4.14 Calibration accuracy. A frequency readout dial shall be provided to determine approximate frequency settings. The dial shall be accurate within ±2.0 MHz measured from the position exhibiting minimum reflected power to the equipment terminal.

3.4.15 Resettability. The ability of the tuning mechanism to reset automatically to a preset frequency shall be such that the change in inherent power insertion loss, after recycling, shall not exceed 0.3 decibel (db) in power (see 4.4.2.9).

3.4.16 Stability. The antenna coupler group shall not require adjustment while operating in varying environmental conditions.

3.4.17 Duty cycle. The antenna coupler group shall be capable of continuous operation.

3.4.18 Insertion loss. Insertion loss between any equipment terminal and the antenna terminal shall be not greater than 2.0 db over the frequency range under normal conditions, and shall not exceed 3 db under extreme environmental conditions (see 4.4.2.5).

3.4.19 Off channel rejection. Minimum attenuation, between the antenna terminal and any equipment terminal, shall be as follows (see 4.4.2.6):

(a) Twenty db at frequencies 1.5 MHz from resonance.
(b) Forty db at frequencies 3 MHz or greater from resonance.

3.4.20 Decoupling. Attenuation between any two equipment terminals shall be a minimum of 60 db if the channels are tuned to frequencies separated by 10 MHz or greater (see 4.4.2.7).

3.4.21 Metering. Four ruggedized meters in accordance with MIL-M-10304 and necessary circuitry shall be supplied to indicate forward and reflected radio frequency power at the equipment terminal of each of the four antenna couplers. The meters shall be capable of continuous monitoring of rf power of each antenna coupler input. The meters and necessary switches or controls shall be located on a separate panel and form a part of the electronic equipment cabinet. The meter calibration shall be as follows:

(a) Forward rf power - 0 to 200 watts.
(b) Reflected rf power - 0 to 200 watts.

3.4.22 Meter accuracy. Forward and reflected rf power shall be indicated within 15 percent of the actual rf power (see 4.4.2.8).

3.4.23 Radio frequency connections. The rf coaxial connectors shall be mechanically and electrically attached to the chassis or frame by sound mechanical means. Loose or press fits, or set screws shall not be relied on to assure mechanical rigidity or electrical continuity.

3.4.24 Antenna connector. The antenna connector shall be the same type that is on the Government furnished equipment.

3.4.25 Equipment channel connectors. The rf connector for each equipment channel shall be the same type that is on the Government furnished equipment.

3.4.26 Grounding. Grounding of the equipment shall be accomplished by means of grounding straps.

3.4.27 Construction. The antenna coupler group shall be of modular design, and shall consist of four antenna couplers, one control-power supply, and one electronic equipment cabinet for mounting the antenna couplers and the control-power supply. The antenna couplers, and the control-power supply shall be removable from the front of the electronic equipment cabinet. The construction shall be such that the AN/SRA-33 closely resembles the Government furnished equipment. All parts shall be electrically and mechanically interchangeable with the Government furnished equipment.

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3.4.28 Mounting. An Electronic Equipment Cabinet, CY-3852/SRA-33, shall be provided to contain the antenna couplers and the control-power supply.

3.4.29 Size. The maximum overall size of the antenna coupler group shall be as follows:

Width - 23 inches.
Height - 53-1/2 inches.
Depth - 34-7/8 inches.

3.4.30 Finish. External finish shall be light gray enamel conforming to type III, class 2 of MIL-E-15090.

3.4.31 Performance. The antenna coupler group shall be capable of operating throughout the frequency range of 225.00 to 399.95 MHz under any of the following conditions:

(a) Antenna terminal connected to any one of the following:
  (1) 50-ohm load.
  (2) Antenna AS-390/SRC.
(b) Each equipment terminal connected to any one of the following:
  (1) 50-ohm load.
  (2) AN/SRC-20, AN/SRC-21, AN/SRC-27 or AN/GRC-27.

3.4.32 Supply voltage and frequency. The antenna coupler group shall be designed to operate from a 115/230 volt ±10 percent 60 hertz (Hz) ±5 percent, single phase, ac power source.

3.4.33 Blown-fuse indicators. Blown-fuse indicators are not required.

3.4.34 Channel lock-out system. A lock-out system shall be provided to prevent damage to receivers and transmitters which are accidentally tuned to the same channel. A switch shall be provided within the control-power supply which will allow the channel lock-out system to be disabled if desired.

3.4.35 Input power. The power input requirements of the antenna coupler group shall not exceed 240 watts during standby, or 600 watts with all four antenna couplers tuning simultaneously, when the primary power source is 115 volts ac, 60 Hz single phase. The power factor shall not be less than 75 percent.

3.4.36 Antenna coupler connectors. Each antenna coupler shall be provided with the following front panel connectors:

(a) AN type receptacle for interconnecting each radio set to the antenna coupler group for preset channel control.
(b) AN type receptacle for interconnecting the antenna coupler to the control-power supply.

3.4.37 Control-power supply connectors. The control power supply shall be provided with the following front panel connectors:

(a) AN type receptacle for connection to 115/230 volts ac, 60 Hz, single phase, power source.
(b) Four AN type receptacles for interconnecting the control-power supply to the four antenna couplers.

3.4.38 Weight. The following maximum weights shall not be exceeded:

	Pounds
(a) Antenna coupler (4 required)	53.0
(b) Control-power supply	52.0
(c) Electronic equipment cabinet	150.0
(d) Special cable, short (2 required)	1.5
(e) Special cable, long (2 required)	1.7
(f) (Total) Antenna coupler group AN/SRA-33	420.4

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3.4.39 Reliability

3.4.39.1 Quantitative reliability requirements. The specified mean-time-between-failure (MTBF) θ_0 (as defined in MIL-STD-781) of this system shall be 2000 hours when it is configured as specified in 3.4.1.

3.4.39.2 Reliability program plan. The supplier shall submit a reliability program plan in accordance with and in sequence with the paragraphs of MIL-STD-785. All paragraphs of MIL-STD-785 shall be addressed and the applicability shall be noted (see 6.2.2).

3.4.39.3 Reliability prediction. The contractor shall perform and prepare the following reliability prediction:

(a) Design prediction based on stress factor and part population in accordance with the requirements of MIL-HDBK-217 (see 6.2.2).

3.4.39.4 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used. When such alternate sources of data are proposed, sufficient information shall be submitted to enable the command to agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.4.39.5 Reliability design guides. The supplier shall make maximum use of the design guides contained in NAVSHIPS 94501, Reliability Design Handbook.

3.4.39.6 Reliability reports. Reports in accordance with MIL-STD-785 that provide a complete accounting of progress on each element defined by the reliability program plan, results achieved and status of actions to resolve problems and correct deficiencies shall be prepared (see 6.2.2).

3.4.39.7 Reliability demonstration tests. The reliability testing shall be in accordance with 4.4.5.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Supplier's quality assurance system. The supplier shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. The quality control system shall be in accordance with MIL-Q-9858. A quality program plan, as it would apply to the requirements of this contract, shall be prepared prior to first article inspection (see 6.2.2). If first article inspection is not required, the quality program plan shall be prepared prior to quality conformance inspection. This program plan shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow-charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the quality control system or the quality program plan shall be submitted to the cognizant Government Quality Assurance Representative for approval prior to use.

4.1.2 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

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4.1.3 Inspection procedures. The supplier shall submit written inspection procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection (see 6.2.2). Each inspection shall be identified in accordance with 4.2 as to classification.

4.1.3.1 Procedures for inspections, tests, demonstrations, and so forth, shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.1.4. Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection
    (1) Inspection covered by subsidiary documents (see 4.4.1)
    (2) Production inspection (see 4.4.2)
    (3) Production control inspection (see 4.4.3)
    (4) Environmental inspection (see 4.4.4)
    (5) Reliability testing (see 4.4.5)
(c) Inspection of preparation for delivery (see 4.4.6)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected
Inspection covered by subsidiary documents		4.4.1	1
Production inspection		4.4.2	1
Production control inspection		4.4.3	1
Environmental inspection		4.4.4	1
Inspection of preparation for delivery		4.4.6	1
Radio frequency connections	3.4.23		1
Antenna connector	3.4.24		1
Equipment channel connectors	3.4.25		1
Grounding	3.4.26		1
Antenna coupler connectors	3.4.36		1
Control power supply connectors	3.4.37		1

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4.3.1 First article inspection report. A first-article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test. The first article inspection report shall indicate specification compliance limits and all inspection results.

4.4 Quality conformance inspection

4.4.1 Inspection covered by subsidiary documents. The following items shall be inspected under the applicable subsidiary documents prior to the inspection of preparation for delivery.

TABLE II. Inspection covered by subsidiary documents.

Item	Requirement paragraph
Metering	3.4.21
Finish	3.4.30

4.4.2 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations approved in accordance with 4.1.3 and shall include the requirements of table III.

TABLE III. Production inspection.

Examination or test	Requirement paragraph	Test method paragraph
(a) Radio frequency power	3.4.3	4.4.2.2
(b) Voltage standing wave ratio	3.4.6	4.4.2.3
(c) Frequency range	3.4.7	4.4.2.4
(d) Insertion loss	3.4.18	4.4.2.5
(e) Tuning time	3.4.13	
(f) Off channel rejection	3.4.19	4.4.2.6
(g) Decoupling	3.4.20	4.4.2.7
(h) Meter accuracy	3.4.22	4.4.2.8
(i) Resetability	3.4.15	4.4.2.9
(j) Controls and control circuitry		
(1) Tuning	3.4.8	
(2) Manual tuning	3.4.9	
(3) Local preset tuning	3.4.10	
(4) Remote preset tuning	3.4.11	
(5) Manual-local-remote switch	3.4.12	
(6) Frequency readout dial accuracy	3.4.14	
(7) Channel lockout system	3.4.34	

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4.4.2.1 Standard test frequencies. When specified for a particular characteristic, the following standard test frequencies shall be utilized.

Channel Number	Channel Frequency (MHz)
1	225.00
2	229.10
3	233.15
4	244.20
5	255.25
6	266.30
7	277.35
8	288.40
9	299.45
10	300.50
11	311.55
12	322.60
13	333.65
14	344.70
15	355.75
16	366.80
17	377.85
18	388.90
19	399.95

4.4.2.2 Radio frequency power. Each AN/SRA-33 shall be operated at full rf power (100 watts nominal input to each channel) for a period of not less than 1 hour. Input and output power shall be recorded throughout the test. The test frequency shall be at the low end of the range. At the end of the 1-hour period, each channel shall be operated for 5 minutes at an intermediate frequency and then for 5 minutes at a frequency at the high end of the range at full power. There shall be no evidence of excessive heating or malfunctioning (see 3.4.3).

4.4.2.3 Voltage standing wave ratio (VSWR). Tests shall be performed at 225.00, 300.00, and 399.90 MHz. Tests shall be performed on each individual antenna coupler (see 3.4.6).

4.4.2.4 Frequency range. Tests shall be performed to show that each channel correctly tunes throughout the frequency range. Tests shall be performed at the standard test frequencies in 4.4.2.1 on each individual antenna coupler (see 3.4.7).

4.4.2.5 Insertion loss. Tests to measure insertion loss shall be made at 225.00, 300.00, and 399.90 MHz. Tests shall be performed on each individual antenna coupler (see 3.4.18).

4.4.2.6 Off channel rejection. Tests shall be performed at 235.00, 342.00, and 385.00 MHz. Tests shall be performed on each individual antenna coupler (see 3.4.19).

4.4.2.7 Decoupling. Tests shall be performed on each individual antenna coupler at 235.00, 342.00, and 385.00 MHz (see 3.4.20).

4.4.2.8 Meter accuracy. Tests shall be performed on each individual antenna coupler at 235.00, 342.00, and 385.00 MHz (see 3.4.22).

4.4.2.9 Resettability. Tests shall be performed on each individual antenna coupler at 225.00, 300.00, and 399.00 MHz (see 3.4.15).

4.4.3 Production control inspection. Production control inspection, including sampling, shall conform to table IV and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved

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in accordance with 4.1.3. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE IV. Production control inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective)
(a) Enclosures	3.3.1.2		6.5
(b) Voltage standing wave ratio	3.4.6	4.4.3.1	6.5
(c) Resetability	3.4.15	4.4.3.1	6.5
(d) Insertion loss	3.4.18	4.4.3.1	6.5
(e) Size	3.4.29		6.5
(f) Supply voltage and frequency	3.4.32		6.5
(g) Input power	3.4.35		6.5
(h) Weight	3.4.38		6.5
(i) Heat test		4.4.3.4	6.5

4.4.3.1 The specified inspection shall be performed at the standard test frequencies specified in 4.4.2.1.

4.4.3.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3.3 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.3.4 Heat test. The equipment shall be operated continuously with rf input power to all four channels as specified in 3.4.3 and with its cooling system in normal operation in an ambient temperature of +50 ±2 degrees centigrade (C) for at least 8 hours without damage or deterioration. There shall be no leakage of any impregnating compounds.

4.4.4 Environmental inspection. Environmental inspection shall be as listed in table V and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be approved in accordance with 4.1.3.

TABLE V. Environmental inspection.

Examination or test	Requirement paragraph
Temperature	3.3.1.1
Shock	3.3.1.3.1
Vibration	3.3.1.3.2
Humidity	3.3.1.4

4.4.4.1 Sampling for environmental inspection. One antenna coupler, AN/SRA-33, shall be selected from each successive 100 units produced or fraction thereof and subjected to the environmental inspection. The first sample unit shall be selected from the first month's production.

4.4.4.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.4, the supplier shall immediately investigate the cause of failure and shall report to the Government Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the Government Quality Assurance Representative does not consider that the corrective action will enable the product to meet specified requirements, or if the supplier cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.4).

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4.4.4.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental inspection may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.4.5 Reliability testing. The reliability of the antenna coupler shall be verified and reported in accordance with MIL-STD-781. The specified MTBF (θ_o) of the system is 2000 hours.

4.4.5.1 Demonstration phase of production reliability tests. The demonstration phase of reliability tests shall be conducted in accordance with test plan IV at Test Level E except that the upper temperature limit shall be +65° C. The sample size shall be five systems selected from the first months production. The reliability requirements shall be successfully demonstrated prior to the shipment of any production equipment.

4.4.5.2 Production acceptance (sampling) phase of production reliability tests. The production acceptance (sampling) phase of reliability tests shall be conducted in accordance with test plan IV-A at Test Level A1, MIL-STD-781. The sample size shall be five systems from each subsequent months production and shall be as specified in 3.4.1. The reliability test shall be successfully completed prior to the shipment of each production lot.

4.4.5.3 Reliability test plan. The supplier shall prepare and submit to the command or agency concerned a detailed reliability test plan in accordance with MIL-STD-785. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the command or agency concerned. No preventative maintenance shall be performed during a reliability test (see 6.2.2).

4.4.5.4 Failure notification. The command or agency concerned shall be notified by telephone or TWX of failures occuring during the tests specified in 4.4.5.1 and 4.4.5.2 that require repair or replacement of functional parts or assemblies. Telephone notification of failure shall be confirmed in writing (see 6.2.2).

4.4.5.5 Failure. Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service. Only relevant failures shall be counted during the reliability test for an accept or reject decision.

4.4.5.6 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.4.5.7 Reliability test reports. The supplier shall submit a report that defines the results of the test at the completion of reliability testing (see 6.2.2).

4.4.6 Inspection of preparation for delivery. Inspections shall be conducted to insure conformance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by suppliers shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.2.1).

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6. NOTES

6.1 Intended use. The equipment in this specification is intended for use in coupling groups of transmitters or combinations thereof to a common antenna for operation in the frequency range 225.00 to 399.95 MHz.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Marking, levels of preservation and packaging and packing required (see 5.1).

6.2.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Reliability program plan (see 3.4.39.2)
(b) Reliability prediction (design) (see 3.4.39.3)
(c) Reliability reports (see 3.4.39.6)
(d) Quality program plan (see 4.1.1)
(e) Inspection procedures (see 4.1.3)
(f) First article inspection report (see 4.3.1)
(g) Report of nonconforming environmental sample units (see 4.4.4.2)
(h) Reliability test plan (see 4.4.5.3)
(i) Confirmation of failure notification (see 4.4.5.4)
(j) Reliability test reports (see 4.4.5.7)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

Preparing activity
Navy - EC

(Project 5985-N341)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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